Logarithm: Short Notes

How to use these notes: read the Formulas box, follow the Steps and Key points, and avoid the Watch out points.

1. Exponential and logarithmic form

Formulas

$$N = a^x \iff \log_a N = x, \qquad N > 0,\ a > 0,\ a \neq 1$$

KEY POINTS

- In $N = a^x$: N is the number, a the base, x the power.
- $a^x = a^y \Rightarrow x = y$ (same base).
- A log is defined only if the number is positive and the base is positive and not 1.

Watch out

Always write the domain conditions before solving.

2. Basic values and the fundamental identity

Formulas

$$\log_a a = 1, \quad \log_a \frac{1}{a} = -1, \quad \log_a 1 = 0, \quad \log_a(a^p) = p, \quad a^{\log_a N} = N$$

KEY POINTS

- $\log_{1/a} a = -1$ as well.
- $a^x = N \Rightarrow x = \log_a N$.

3. Properties of logarithm

Formulas

$$\log_a m + \log_a n = \log_a(mn), \qquad \log_a m - \log_a n = \log_a \frac{m}{n}$$

$$\log_a(m^p) = p \log_a m, \qquad \log_{a^p} m = \frac{1}{p} \log_a m, \qquad \log_{1/a} m = -\log_a m$$

KEY POINTS

- All terms must have the same base before clubbing.
- Even power: $\log_a m^2 = 2 \log_a |m|$.
- Clubbing and splitting change the domain; recheck it.

Watch out

$\log(m + n)$ is not $\log m + \log n$.

4. Base changing theorem

Formulas

$$\log_a b = \frac{\log_c b}{\log_c a}, \qquad \log_c a = (\log_b a)(\log_c b), \qquad \log_a b = \frac{1}{\log_b a}, \qquad a^{\log_c b} = b^{\log_c a}$$

KEY POINTS

- A chain $\log_a b \cdot \log_b c \cdot \log_c d = \log_a d$ collapses.
- Positive value: number and base on the same side of 1.
- Negative value: number and base on different sides of 1.

5. Antilog, characteristic, mantissa, digits, zeroes

Formulas

$$\text{antilog}_a(x) = a^x, \qquad \log_{10} N = c + m,\ c \in \mathbb{Z},\ 0 \le m < 1$$
$$N = p \times 10^n\ (1 \le p < 10), \qquad \text{digits} = c + 1, \qquad \text{zeroes after decimal} = |c + 1|$$

KEY POINTS

- c = characteristic (integer part), m = mantissa (never negative).
- For a negative log value, c is the integer just below it.
- Digits formula is for $N \ge 1$; zeroes formula is for $0 < N < 1$.

Watch out

Mantissa of -15.05 is 0.95 (with $c = -16$), not 0.05.

6. Log equations: Type 1 and Type 2

Formulas

$$\log_a m = p \Rightarrow m = a^p, \qquad \log_a m = \log_a n \Rightarrow m = n$$

STEPS

1. Write the domain: every number > 0, every base > 0 and $\neq 1$.
2. Club the logs to one side, or put $t = \log_a x$ (or $t = a^x$) when the log is repeated or squared.
3. Remove the log (or solve in t), return to x.
4. Reject the roots outside the domain.

Watch out

$a^x = t$ gives $t > 0$; reject non-positive t.

7. Log equations: Type 3 (variable power)

Formulas

$$f^g = h \Rightarrow g \log f = \log h, \qquad a^b = a^c \Rightarrow a = 1 \text{ or } b = c$$

STEPS

1. Make sure $f > 0$.
2. Take log (base 10 or a convenient base) of both sides.
3. Put $t = \log x$ and solve; for $a^b = a^c$ also test $a = 1$.

8. Logarithmic inequalities

Formulas

$$a > 1:\ \log_a M > \log_a N \iff M > N > 0; \qquad 0 < a < 1:\ \log_a M > \log_a N \iff 0 < M < N$$
$$a > 1:\ \log_a M > p \iff M > a^p; \qquad 0 < a < 1:\ \log_a M > p \iff 0 < M < a^p$$

KEY POINTS

- Base > 1: inequality stays; base in $(0, 1)$: it reverses.
- Always add 'number > 0'.
- Variable base: two cases, then union.

9. Modulus function and properties

Formulas

$$|x| = \begin{cases} x, & x \ge 0 \\ -x, & x < 0 \end{cases} \qquad |x| = \sqrt{x^2}, \qquad |x|^2 = x^2$$

$$|x| \ge 0, \quad |x| = |-x|, \quad |xy| = |x||y|, \quad |x+y| = |x| + |y| \iff xy \ge 0, \quad |x-y| = |x| + |y| \iff xy \le 0$$

KEY POINTS

- $|f(x)|$ is defined by splitting at the zeros of f.
- $\sqrt{x^2} = |x|$, not x.

10. Modulus equations

Formulas

$$|f| = k \ (k \geq 0) \Rightarrow f = \pm k; \qquad |f| = |g| \Rightarrow f^2 = g^2; \qquad |f| = g \Rightarrow g \geq 0, \ f = \pm g$$

STEPS

1. $|f| = k$: if $k < 0$ there is no solution; else $f = \pm k$.
2. $|f| = |g|$: square both sides (no extra roots).
3. $|f| = g$ or several moduli: define the modulus on each interval, solve, and keep roots lying in their interval.

Watch out

After squaring $|f| = g$, check $g \geq 0$ for every root.

11. Modulus inequalities

Formulas

$$|f| < k \iff -k < f < k; \qquad |f| > k \iff f > k \text{ or } f < -k \qquad (k > 0)$$
$$|f| < |g| \iff f^2 < g^2 \iff (f-g)(f+g) < 0$$

KEY POINTS

- Negative constant: $|f| < k$ has no solution; $|f| > k$ holds for every x in the domain.
- Square only when both sides are ≥ 0.
- Several moduli: split at the zeros, solve each interval, union the pieces.

12. Exponential inequalities

Formulas

$$a > 1: \ a^x > a^y \iff x > y; \qquad 0 < a < 1: \ a^x > a^y \iff x < y$$

KEY POINTS

- Bring both sides to the same base first.

13. Irrational inequalities

Formulas

$$\sqrt{f} < g \iff f \geq 0, \ g > 0, \ f < g^2$$
$$\sqrt{f} > g \iff [f \geq 0, \ g < 0] \text{ or } [g \geq 0, \ f > g^2]$$

KEY POINTS

- $\sqrt{f} \geq 0$ wherever it is defined.
- Take the union of the two cases for $\sqrt{f} > g$.

14. Comparing logs and nested radicals

KEY POINTS

- Compare two logs: same base, or compare each with 1 using the sign rule.
- $t + \dfrac{c}{t} \geq 2\sqrt{c}$ for $t > 0$ (AM-GM) when a log and its reciprocal occur.
- $\sqrt{a \pm 2\sqrt{b}} = \sqrt{p} \pm \sqrt{q}$ where $p + q = a, \ pq = b$.

15. Common mistakes checklist

STEPS

1. Forgot the domain: number > 0, base > 0 and $\neq 1$.
2. Did not reject extraneous roots after clubbing logs or squaring.
3. Reversed the inequality only for base in $(0, 1)$.
4. Wrote $\sqrt{x^2} = x$ instead of $|x|$.
5. Took $a^x = t$ with $t \leq 0$.
6. Mantissa written negative.
7. Squared an inequality whose sides can be negative.